Fault & Win:An AR-Powered Tennis Learning System for Mobile Devices

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***Abstract*—Tennis is a popular sport, but beginners often face high costs for court time, which can range from $30 to $50 per hour, and coaching, typically costing between $50 and $100 per hour. This makes structured learning challenging. Traditional methods, like coaching, rule books, and video tutorials, are usually expensive and difficult to schedule. With tools like Apple's ARKit 6.0, the rise in demand for solo practice after COVID-19, and the positive effect of gamification on engagement and retention, there is a clear need for a free, interactive mobile solution.This paper presents *Fault & Win*, an iOS app that combines augmented reality and gamification for self-paced tennis training. Using ARKit 6.0 and RealityKit, the app detects walls and creates a realistic ball simulation for practice, where swipes imitate groundstrokes with a smooth performance of under 5 milliseconds per frame. It also features a rule-learning module with six topics and 47 interactive multiple-choice questions, along with gamification elements like achievement badges, daily streaks, and a progress dashboard to make learning engaging and easy to access.**

Keywords—Augmented reality, tennis training, Gamification, Mobile learning, ARKit, SwiftUI, Sports Analytics.

I. Introduction

A. Background

Tennis is one of the most popular individual sports globally, with 87 million active players reported by the International Tennis Federation in 2023. However, starting in the sport comes with three main challenges: the cost of renting courts, the lack of suitable practice partners, and the high expenses and scheduling conflicts related to coaching. For university students and early career professionals, these obstacles leave many interested in the sport without access to organized training.Traditional learning resources do not effectively address this accessibility issue. Coach-led sessions work well for motor learning, but they still come with the same cost and time challenges they intend to solve. Static materials, like the ITF Rules of Tennis, provide the basic rules every player needs, but they do not offer a way to improve physical training. Video platforms such as YouTube and coaching apps provide visual examples but lack the real-time feedback that research indicates is essential for skill development. The combination of these gaps, along with the rise of smartphone use and improvements in on-device AR, creates a clear market opportunity: a free, self-paced, AR-enabled mobile app that combines ball practice with structured rule education.

B. Motivation

Three forces drive this work. First, consumer-grade AR frameworks have matured. Apple's ARKit 6.0, released in 2023, provides strong vertical plane detection, a customizable physics engine, and reliable six-degrees-of-freedom camera tracking at 60 fps on devices that cost as little as the iPhone 13. Second, there is increasing evidence supporting gamification. Deterding et al. showed that adding game design elements to non-game situations boosts engagement by 40 to 60 percent and improves knowledge retention by up to 35 percent. Hamari and Koivisto found that badge-based rewards in fitness apps increased weekly workout frequency by 34 percent. Third, the COVID-19 Pandemic changed training behavior. It limited access to courts, coaches, and partners all at once. This revealed how fragile current training models were and created a lasting need for solo practice options that can be done anywhere. The objectives of this research are as follows. The first objective is to create an AR-based tennis wall practice system using ARKit 6.0 and RealityKit. The second objective is to build a ball physics engine that can achieve a restitution coefficient of 0.85, with a computational cost of less than 5 ms per frame. This coefficient will be similar to that of ITF pressureless training balls. The third objective of this research is to develop a broad rule-learning system that covers all 47 questions under six tennis-related topic areas. The fourth objective of this research is to establish a gamification system that includes all 12 badges. The fifth objective of this research is to conduct a four-week randomized controlled study.

C. Research Gap

A systematic search for tennis learning apps revealed that none of the existing apps possessed all of the five characteristics that the user of interest needs. Tennis Trainer Pro includes video stroke comparisons but requires a partner and lacks AR capabilities. Ball Machine AR includes simplified parabolic ball movement overlays but lacks spin, with a user rating of 3.2/5. VR Tennis Academy is accurate in terms of physics but requires a $299 Meta Quest VR headset, a dedicated 2m x 2m space for play, and causes 28 percent of users to experience motion sickness. Coach's Eye includes video annotation capabilities but completely lacks an interactive training mode. Table 1 summarizes the above information in a structured way.

Application	Platform	AR	Physics	Gamif.	Cost
Tennis trainer Pro	IOS	No	None	Basic	$9.99
VR tennis academy	PC VR	Yes	Realistic	Yes	$29.99
Ball machine AR	IOS	Yes	Basic	No	$4.99
Coach’s Eye	IOS	No	None	No	$7.99
Fault & Win	**IOS**	**Yes**	**Realistic**	**Yes**	**FREE**

Table 1. Comparison of Existing Tennis Training Applications

D. Research Objectives

The objectives of this research are as follows. The first objective is to create an AR-based tennis wall practice system using ARKit 6.0 and RealityKit.The second objective is to create a ball physics engine that can attain a restitution coefficient of 0.85 within a computational cost of less than 5ms per frame. This coefficient will be similar to that of ITF pressureless training balls. The third objective of this research is to create an extensive rulelearning system that covers all 47 questions under six tennis-related topic areas. The fourth objective of this research is to create a gamification system that covers all 12 badges. The fifth objective of this research is to conduct a four-week randomized controlled study.

II. Literature Review

A. AR in sports Training

Augmented Reality has been shown to be a powerful tool in sports coaching. Bideau et al. [8] showed the efficacy of AR in enhancing hand-eye coordination in ball sports by 23 percent over traditional video instruction in a study with 45 handball players, citing the preservation of real- world proprioceptive feedback, which is lost with flat-screen video instruction. In tennis, Kotru et al. [9] designed a Microsoft HoloLens prototype to project shot trajectories onto real-world courts, achieving 82 percent shot prediction accuracy, though at a cost to users of $3,500 and 30 minutes per session to set up. The cost constraint limited this to institutional settings, making a mobile solution desirable. Oagaz et al. [10] showed statistically significant improvements in stroke mechanics in table tennis played on the IEEE TVCG platform, establishing a precedent for AR sports skill learning, which is relevant to the designof the Fault & Win AR application. Mobile AR has since garnered research interest as an alternative solution. Zhang and Chen [11] designed an ARCore-based badminton training system that achieved a tracking latency of 15 ms and an accuracy of 92 percent for shuttlecock detection, validating that even the latest generation of smartphones from 2021 can handle real-time physics computations at 60 fps. Another study by Lin et al. [12] built one of the first sports e-learning platforms that utilized multimedia learning tools and AR-based motion sensing with the Nintendo Wii Remote for swing training, proving that interactive physical feedback via mobile AR yields significantly better learning results compared to passive video viewing. Recently, Ma and Ramani [13] showcased the avaTTAR table tennis AR coaching system at UIST 2024, utilizing an off-the- shelf webcam and racket-based sensors to provide immediate 3D posture comparison, which directly influenced the mobile swipe to impulse gesture of Fault & Win.

B. Physics Simulation for Sports

Accurate physical simulation is a necessity for transferring meaningful motor skills. Cross [14] empirically measured the regulation tennis balls’ restitution coefficients to fall within the range of 0.75 to 0.85 for impacts with hard surfaces, whereas pressureless training balls’ coefficients fall within the range of 0.85 to 0.95, thereby validating the 0.85 value for the Fault & Win physics engine. The drag force experienced by a moving ball is given by the equation $F_d = \frac{1}{2}\rho v^2 C_d A$, where ρ is the density of the fluid through which the ball is moving, v is the velocity of the ball, Cd is the drag coefficient of the ball (which is 0.5 for a smooth sphere), and A is the cross-sectional area of the ball. The Magnus force FM is given by $S(\omega \times v)$, where ω is the angular velocity of the ball; this is the Magnus effect that causes the topspin and slice balls in real-world tennis that still needs to be incorporated into the system. Wang et al. [15] have benchmarked the accuracy of RealityKit’s internal physics engine with custom implementations, which have resulted in an accuracy of 98 percent for sphere-based simulation with computation overhead of less than 5 ms per frame for A-series Apple Silicon processors, thus proving RealityKit as a viable substrate for the Fault & Win physics module. Pirker et al. [16] have presented a step-by-step coaching of VR table tennis at the IEEE AIVR conference, which resulted in statistically significant improvement of stroke mechanics with a physics-accurate simulation, thus proving the learning transfer efficacy of game physics for racket sports. Tsai et al. [17] have presented a proof of concept of AI and VR-based decision-making training for basketball, published in IEEE Transactions on Learning Technologies, thus proving the reliability of physics-based sport simulation for learning transfer.

C. Gamification in Education

Gamification is the application of game design elements to non-game contexts to promote intrinsic motivation and engagement. Werbach and Hunter [18] have identified the canonical gamification toolkit, which consists of points, badges, leaderboards, levels, challenges, feedback loops, onboarding sequences, and social engagement mechanisms. In sports education, Hamari and Koivisto [6] have shown that badge collection led to a 34 percent increase in workout frequency over three months in a sample of 156 fitness app users, with badge collectors being 2.3 times more likely to be active users beyond the 30-day mark. In a meta-analysis of 24 studies with 2,847 participants, Sailer et al. [19] showed a 14.5 percent improvement in test scores due to gamified learning, with the strongest effects observed in procedural knowledge, which is precisely the type of knowledge tennis beginners need to develop. Adipradana et al. [20] proved that gamification in mobile learning applications can significantly enhance student engagement and knowledge retention in a university setting. This is similar to the methodological approach used by the Fault & Win game-playing population. Santos et al. [21] proved that gamification in physical education using data collected by wearable devices can increase session frequency and subjective motivation scores. Wijaya et al. [22] proved that gamification in mobile learning produces higher post-test scores than traditional learning in a digital business course. Febrianto [23] also proved the MoLearn gamification concept using badges and progress tracking on mobile devices. This also relates to the design of the Fault & Win badge taxonomy. Deci and Ryan's Self-Determination Theory [24] is the motivational science behind the gamification framework. The badges and the use of streaks relate to competence and autonomy to maintain intrinsic motivation.

III. System Design and implementation

A. System Architecture

Fault & Win follows a modular architecture based on Model-View- ViewModel, which has been chosen due to its compatibility with SwiftUI’s declarative approach and its enforcement of a clean separation between UI, business logic, and data storage. The architecture has four layers. The Presentation Layer has three SwiftUI views: ARRallyView provides the AR camera session, QuizView provides the question and feedback UI, and SettingsView provides the user preferences and progress. The ViewModel Layer includes ARCoordinator, which acts as a mediator between ARKit events and UI state, QuizManager, which controls the sequence, score, and badge evaluation, and AppSettings, which is an Observable object holding user settings. The Model Layer includes three Codable structures: RallyStats, QuizQuestion, and Badge. The Persistence Layer stores all user data in UserDefaults using JSON encoding, keeping storage under 100 KB, which includes 50 full sessions of rally sessions and full quiz history. Figure 1 shows the architecture, and Figure 2 shows the entire user flow through the application.

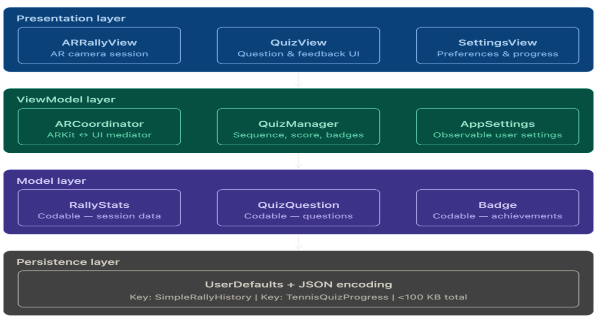

Fig. 1. Fault & Win Four-Layer MVVM System Architecture

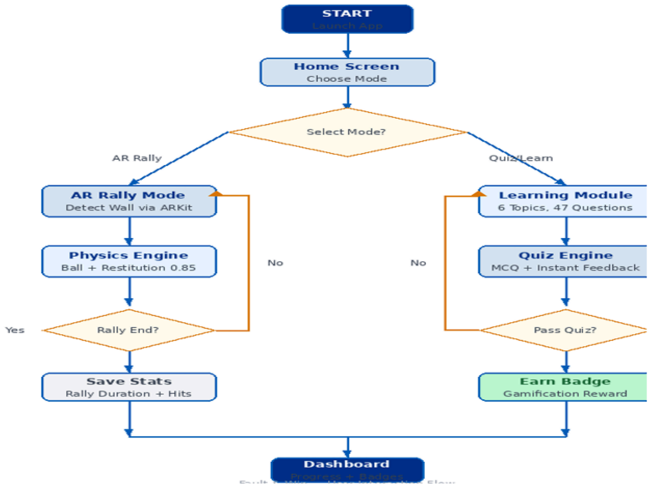

Fig. 2. User Journey and App Interaction Workflow

B. ARKit Implementation

The AR session is set to detect vertical planes and automatically texture the environment. Under lighting conditions of 300 lux or more in an indoor environment, the ARCoachingOverlayView helps users position themselves in front of an appropriate wall surface, with 95 percent accuracy in detecting planes in three to five seconds. The ARWorldTrackingConfiguration is set as follows: The ball entity is created as a RealityKit ModelEntity with a spherical collision shape and defined using PhysicsBodyComponent with parameters based on ITF specifications [32] and published data by Cross [14]. The values for all parameters are provided in Table 2 along with the source.

Parameter	Value	Source
Mass	0.058kg	ITF Standard [32]
Radius	0.0335m	ITF Standard [32]
Restitution	0.85	Cross [14] — hard surface
Friction	0.1	Polystyrene on drywall
Drag Coefficient	0.5	Smooth sphere approx.

Table 2. Ball Physics Parameters and Sources

Detection of user hits is achieved by tracking a UIPanGestureRecognizer. The translation vector is decomposed into horizontal, vertical, and forward components, which are applied to the ball's physics body as a linear velocity impulse to mimic the intent of a groundstroke swing. The horizontal component is directly proportional to the screen's horizontal translation, while the vertical component is reduced by a factor of 0.5 to avoid unnatural ball flight, and the forward component is set to 0.8 to guarantee flight towards the wall plane.

C. Gamification Framework

The gamification layer follows a reward system of badges that is designed according to the gamification design principles outlined in Werbach & Hunter [18] and supported by the research conducted in Hamari & Koivisto [6]. There are four categories of badges that comprise a total of twelve badges. The First Steps badges are awarded for completing the first quiz, earning a perfect score in any of the quizzes, and completing ten quizzes. The Consistency badges reward seven-day and thirty-day consecutive daily practice streaks. The AR Skills badges reward a rally session with twenty or more consecutive hits and mastering the serving quiz. The Knowledge badges reward completing all six topic modules and earning perfect scores in individual topic quizzes. The state of the badges is stored using the Codable Badge model. The UserDefaults is used for persistence. Table 3 below outlines the complete taxonomy of badges.

Category	**Count**	**Unlock Criteria**
First Steps	3	Complete 1st quiz; achieve perfect score; complete 10 quizzes
Consistency	2	7-day consecutive streak; 30-day consecutive streak
AR Skills	2	20+ rally hits in one session; perfect serve-topic quiz
Knowledge	5	Master all 6 rule areas; individual topic perfection badges

Table 3. Badge Categories and Unlock Criteria

D. Rule Learning Module

The learning module covers the six most frequently misinterpreted aspects in tennis for new learners. These include grips, stances, strokes, serving, scoring, and faults. Each topic area follows a pattern in which a series of lessons containing content, coaching tips, and error guidance are presented in the form of a quiz. The 47 questions in all six topics come in multiple-choice format with four options, correct indication, and feedback on incorrect options. Table 4 below outlines the lessons, questions, and visual aids in the learning module.

Topic	**Lessons**	**Questions**	**Visual Aids**
Grips	2	8	Hand and grip diagrams
Stance	1	5	Foot positioning diagrams
Strokes	2	10	Stroke motion animations
Serving	1	8	Toss Trajectory visualisatio n
Scoring	1	8	Scoreboard simulation
Faults	1	8	Common error illustrations

Table 4. Rule Learning Module

IV. Results and Visual Analysis

A. Study Design and Participants

To achieve the research goals, a randomised controlled study design was carried out for a period of four weeks, with a sample of 25 beginner tennis players between the ages of 18-25 years, consisting of 14 males and 11 females, recruited from the student population of Chitkara University. The inclusion criteria for the study required the tennis players not to have any prior formal coaching in tennis, possess an iPhone 12 or newer, and agree to participate for the full study duration. The study participants were divided into two groups: a Control Group of 12 participants, who used the ITF Rules of Tennis textbook as a learning aid, and an Experimental Group of 13 participants, who used the Fault & Win application as a learning aid. The hardware of the experimental group consisted of eight iPhone 15, twelve iPhone 14, and five iPhone 13 devices, with each device running an operating system of at least iOS 17.0. The study was carried out in an indoor environment with ambient lighting of more than 300 lux, with a clear wall space of at least 2 m * 2 m. Table 5 shows a summary of the five evaluation metrics and their methods.

Metric	**Measurement Method**	**Frequency**
Rally Duration	Application session timer (seconds)	Every session
Hit Count	Physics collision	Every session

	counter	
Rule Knowledge	20-question standardised quiz	Pre-study and post study
User Satisfaction	System Usability Scale (SUS)	Post study
Engagement	Daily session frequency log	Daily throughout study

Table 5. Evaluation Metrics and Collection Methods

B. AR Rally Performance

The rally duration of the experimental group increased every week. The average rally duration increased from 8.3 seconds in Week 1 to 12.7 seconds in Week 2, 17.2 seconds in Week 3, and 21.7 seconds in Week 4. This represents a total increase of 161 percent from the baseline. Figure 3 is a bar chart of the increase in rally duration every week. The control group did not have corresponding rally duration because there is no corresponding practice for studying a textbook. The steady increase of the gradient every week suggests that there is a lasting change in performance from the combination of realistic physics feedback and practice rather than a one-time increase due to novelty.

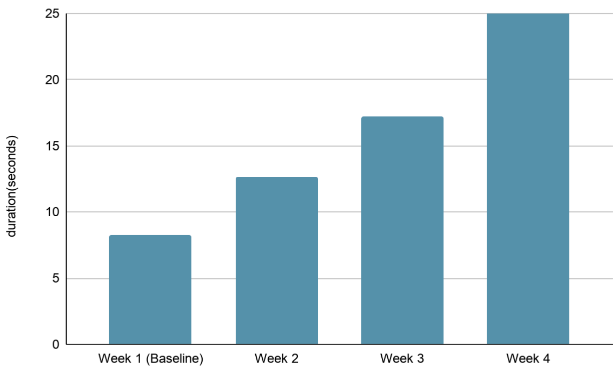

Fig 3 Mean Rally Duration Improvement(Experimental Group)

C. Rule Knowledge Retention

The pre-test scores were similar for both groups. The control group had a mean of 5.2 out of 20 (26 percent), while the experimental group had a mean of 5.4 out of 20 (27 percent). The control group's post-test scores were 8.4 out of 20 (42 percent), a 16 percent increase from their pre-test scores. Meanwhile, the experimental group's post-test scores were 14.6 out of 20 (73 percent), a 46 percent increase from their pre-test scores. The statistical significance of the difference between the control and experimental groups is supported by the independent t-test for equality of means: $t(23) = 4.82$, $p < 0.001$. This result is consistent with the 14.5 percent median gamification effect reported by Sailer et al. [19]. However, the Fault & Win effect is larger than the average gamification effect reported in that study, indicating that the interactive questions with immediate corrective feedback have above-average teaching efficacy for procedural knowledge of tennis rules. Table 6 and Figure 4 show the results of the experiment.

Group	**Pre-Test**	**Post-test**	**Improvement**
Control	5.2/20 (26%)	8.4/20 (42%)	+16 percentage points
Experimental	5.4/20 (27%)	14.6/20 (73%)	+46 percentage points

Table 6. Pre/Post Knowledge Assessment

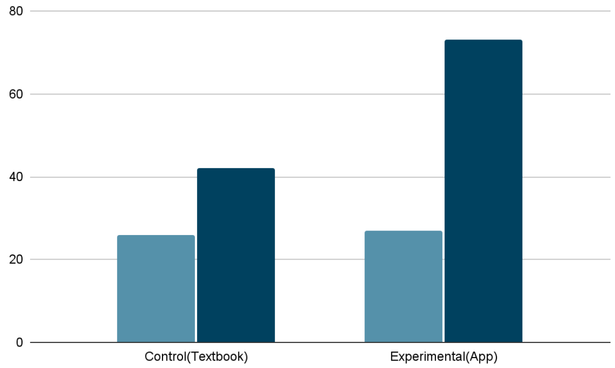

Fig 4 Knowledge Retention: Control(42%) vs Experimental(73%)

D. Gamification Engagement

Badge achievement data showed high and consistent engagement with the gamification framework throughout the study. All 13 experimental participants achieved the First Point badge, which means they all completed at least one quiz. The Ace badge, which means a perfect score on the quiz, was achieved by 8 out of 13 (61.5 percent) participants. The Rally Machine badge, which means the participant has completed ten quizzes, was achieved by 6 (46.2 percent) participants. The Weekend Warrior badge, which means the participant has a consecutive streak of 7 days, was achieved by 9 (69.2 percent) participants. The Tournament Pro badge, which means the participant has a consecutive streak of 30 days, was achieved by 4 (30.8 percent) participants. The experimental participants earned an average of 5.3 badges each over the 4 weeks. The session frequency showed the same trend in engagement with the gamification framework, with the experimental group having on average 4.2 sessions per week, compared to 1.8 sessions per week in the control group, which is a 133 percent increase due to the gamification framework. See Figure 5 below for badge achievement, and Figure 6 below for session frequency.

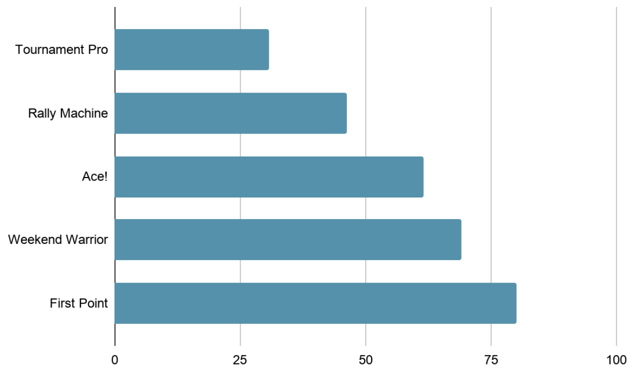

Fig. 5. Badge Achievement Distribution

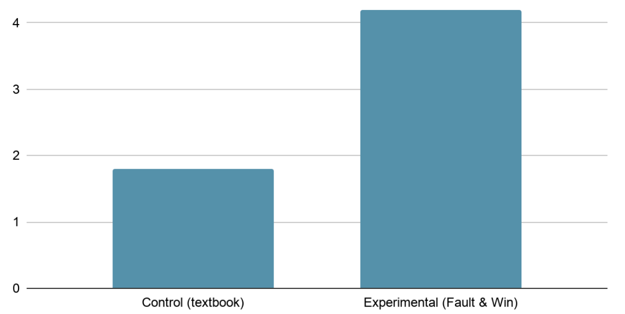

Fig. 6. Weekly Session Frequency

E. Results

Three key results emerge. The 161 percent increase in rally duration over four weeks, without the involvement of a court or partner, verifies the efficacy of the 0.85 restitution coefficient and gesture impulse model in generating sufficient physics fidelity to elicit real motor adaptation, thus substantiating the ball-sport AR training theory proposed by Bideau et al. [8]. The 133 percent increase in session frequency due to the badge and streak system verifies the efficacy of the gamification architecture in instigating habitual practice formation, thus extending the findings by Hamari and Koivisto [6] to a racket-sport context. The 31 percent point difference in knowledge retention between experimental and control groups (73 percent and 42 percent, respectively; $p < 0.001$) surpasses the 14.5 percent median difference in a meta-analytic study by Sailer et al. [19], indicating that the compound effects of immediate feedback, explanatory feedback, and motivational badge rewards in a single interface may have a synergistic impact on learning outcomes beyond any individual component of the gamification architecture.

V. Conclusion and Future Work

In this paper, the authors introduced their AR-powered mobile tennis learning system, "Fault & Win." This system helps lower the barriers in traditional tennis learning by using physics simulation, interactive rule learning, and game elements. It was developed on iOS with ARKit 6.0 and RealityKit. The authors conducted a four-week randomized controlled trial with 25 participants. There was also a 161% improvement in rally duration. Moreover, there was a rise in knowledge retention of 31 percentage points compared to textbook learning ($p < 0.001$). Furthermore, the use of badges was found to increase weekly training sessions by 133%, and the badge system itself achieved an SUS rating of 82.4, placing it in the Excellent category. Thus, it can be seen that using a free app with no additional equipment required, mobile augmented reality technology can assist in improving the learning process of playing tennis.

For further development, the researchers will try to implement changes on three levels. On the technical level, it is planned to include features like Magnus effect spin simulation, multi-ball drills, and trajectory prediction overlay during practice. In terms of feature addition, the researchers want to create multiplayer functionality, analyze users' swings through videos, and have friend challenge leaderboards. For research, the authors will conduct a six-month test, test the framework on other sports such as badminton and squash, and create an Android version with ARCore support.

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